

ASSESSMENT TOOL – 2

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO5	Assessment	Pseudocode Writing
Weightage		Total Marks	

Student Name	Cindy R	Reg. No.	192412571
Section / Batch	ECE-2024	Date of Submission	19/06/2026
Faculty Incharge	Dr R S Selvan	Assessment	Individual Submission

Total Marks		Obtained Marks	
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Code of Conduct:

20
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I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Network Throughput and Packet Loss

Algorithm Network Performance

BEGIN

totalDataReceived = 8000

time = 4

packetsSent = 200

packetsReceived = 100

IF time = 0 THEN

DISPLAY "Error: Time cannot be zero".

STOP

END IF

IF packetsSent = 0 THEN

DISPLAY "Error: Total packets sent cannot be zero".

STOP

END IF

throughput = totalDataReceived / time

packetsLost = packetsSent - packetsReceived

packetLossPercentage = (packetsLost / packetsSent) * 100

DISPLAY "Network performance summary".

DISPLAY "Total Data Received: ", totalDataReceived, "bits"

DISPLAY "Time: ", time, "seconds"

IF Throughput > 1500 AND PACKETLOSSPERCENTAGE < 10 THEN

networkStatus ← "Efficient"

ELSE

networkStatus ← "Inefficient"

END IF

DISPLAY "Network Classification: ", networkStatus

END

Result:

$$\text{Throughput} = 8000/4 = 2000 \text{ bps}$$

$$\text{Packet loss} = 200 - 180 = 20 \text{ packets}$$

$$\text{Packet loss} = (20/200) \times 100 = 10\%$$

$$\text{Throughput} > 1500 \text{ bps} \rightarrow \text{Yes}$$

$$\text{Packet loss} < 10\% \rightarrow \text{No because it's exactly } 10\%$$

Therefore:

Network classification = Inefficient.

2. Bandwidth Utilization Monitoring

Algorithm Bandwidth Monitoring

BEGIN

links $\leftarrow$ ["Link1", "Link2", "Link3", "Link4"]

number of links $\leftarrow$ LENGTH (links)

While True Do

DISPLAY "starting bandwidth monitoring..."

FOR $i \leftarrow 0$ TO number of links - 1 DO

utilization[i] $\leftarrow$ GET_BANDWIDTH_UTILIZATION (links[i])

DISPLAY links[i], "Utilization:", utilization[i], "%"

IF utilization[i] > 80 THEN

DISPLAY "Warning: ", links[i], " is overloaded".

DISPLAY "Utilization = ", utilization[i], "%"

DISPLAY "Recommendation: Switch traffic to backup

link

SWITCH-TRAFFIC-TO-BACKUP (links[i])

ELSE

FOR $i \leftarrow 0$ TO number of links - 1 DO

IF status $[i]$ = "Failed" THEN

score $[i] \rightarrow 0$

ELSE

Bandwidthscore $\leftarrow$ Normalize (bandwidth $[i]$)

delay score $\leftarrow 1 / (1 + \text{delay } [i])$

utilizationscore $\leftarrow 1 - (\text{utilization } [i] / 100)$

score $[i] \leftarrow$

$(0.5 \times \text{bandwidthscore}) +$

$(0.3 \times \text{delay score}) +$

$(0.2 \times \text{utilizationscore})$

END IF

END FOR

bestscore $\leftarrow -1$

best path $\leftarrow$ NONE

FOR $i \leftarrow 0$ TO number of links - 1 DO

IF status $[i]$ = "Failed" THEN

IF score $[i] >$ bestscore THEN

bestscore $\leftarrow$ score $[i]$

bestpath $\leftarrow$ links $[i]$

END IF

END IF

IF BESTPATH $\neq$ NONE THEN

SELECT PATH (bestpath)

ENDIF

WAIT 10 seconds

DISPLAY links[i], " is operating completed",

WAIT 10 seconds

END WHILE

END

Conclusion:

The algorithm continuously monitors the utilization of every network link and stores the measured values in the utilization[] array. If utilization exceeds 80%, the link is considered overloaded. The algorithm then recommends or automatically switches traffic to a backup link. This helps in efficient bandwidth management by distributing traffic before a link becomes critically congested.

3 Dynamic Best-Path Selection

Algorithm Dynamic Best Path Selection

BEGIN

links ← ["Link1", "Link2", "Link3", "Link4", "Link5",
"Link6"]

number of links ← LENGTH(links)

WHILE TRUE DO

DISPLAY "Collecting network information..."

FOR i ← 0 TO number of links - 1 DO

bandwidth[i] ← GET_BANDWIDTH(links[i])

delay[i] ← GET_DELAY(links[i])

END FOR

END WHILE

END

Conclusion:

- * Higher Bandwidth $\rightarrow$ Better score
- * Lower delay $\rightarrow$ better score
- * Lower utilization $\rightarrow$ better score
- * Failed of links $\rightarrow$ score of zero

Benefits:

- * Network reliability
- * Routing efficiency
- * Fault tolerance
- * Congestion prevention
- * Dynamic adaptation

